CARBON FOOTPRINT MONITORING SYSTEM FOR SMART CAMPUSES USING IoT AND OCCUPANCY ANALYTICS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The increasing demand for electricity in educational institutions and large campus infrastructures has led to a significant rise in carbon emissions associated with building operations. Universities today function as small-scale urban environments consisting of classrooms, laboratories, administrative blocks, residential facilities, and shared services, all of which contribute collectively to energy consumption patterns. However, conventional monitoring approaches used in campus environments primarily focus on aggregate electricity usage reporting rather than intelligent analysis of consumption behavior. As a result, opportunities for optimizing energy utilization and reducing carbon footprint often remain unidentified. The development of data-driven monitoring frameworks capable of supporting sustainable infrastructure management has therefore become an important requirement for modern smart campuses.

This project presents a carbon footprint monitoring framework that integrates Internet of Things (IoT) based sensing concepts with occupancy-aware analytics to estimate, analyze, and optimize electricity consumption patterns within campus buildings. The proposed system adopts a structured methodology that combines time-series energy usage evaluation with threshold-based demand-side management strategies to identify peak-load conditions and simulate emission reduction possibilities. Instead of relying solely on static consumption records, the framework introduces intelligent analysis capable of supporting proactive decision-making for campus administrators.

The system architecture is designed as a multi-layer analytical pipeline consisting of data acquisition, preprocessing, consumption modeling, emission estimation, and decision-support modules. Building-level electricity consumption datasets similar to benchmark smart-building energy datasets are utilized to simulate real-world campus energy behavior. These datasets are processed to identify temporal variations in usage across different hours of the day, working schedules, and occupancy conditions. The integration of occupancy-aware threshold logic enables the framework to estimate avoidable energy consumption during underutilized periods such as evenings, weekends, and non-instructional intervals.

Carbon emission estimation within the proposed system is performed using standardized electricity-to-emission conversion factors derived from accepted environmental accounting methodologies. By combining consumption analysis with emission modeling, the framework provides both energy-level and carbon-level insights that support sustainability-oriented infrastructure planning. The introduction of threshold-based optimization further enables simulation of multiple reduction scenarios in which peak demand loads are shifted or minimized without affecting operational requirements of campus facilities.

The analytical results obtained from the proposed framework demonstrate that occupancy-

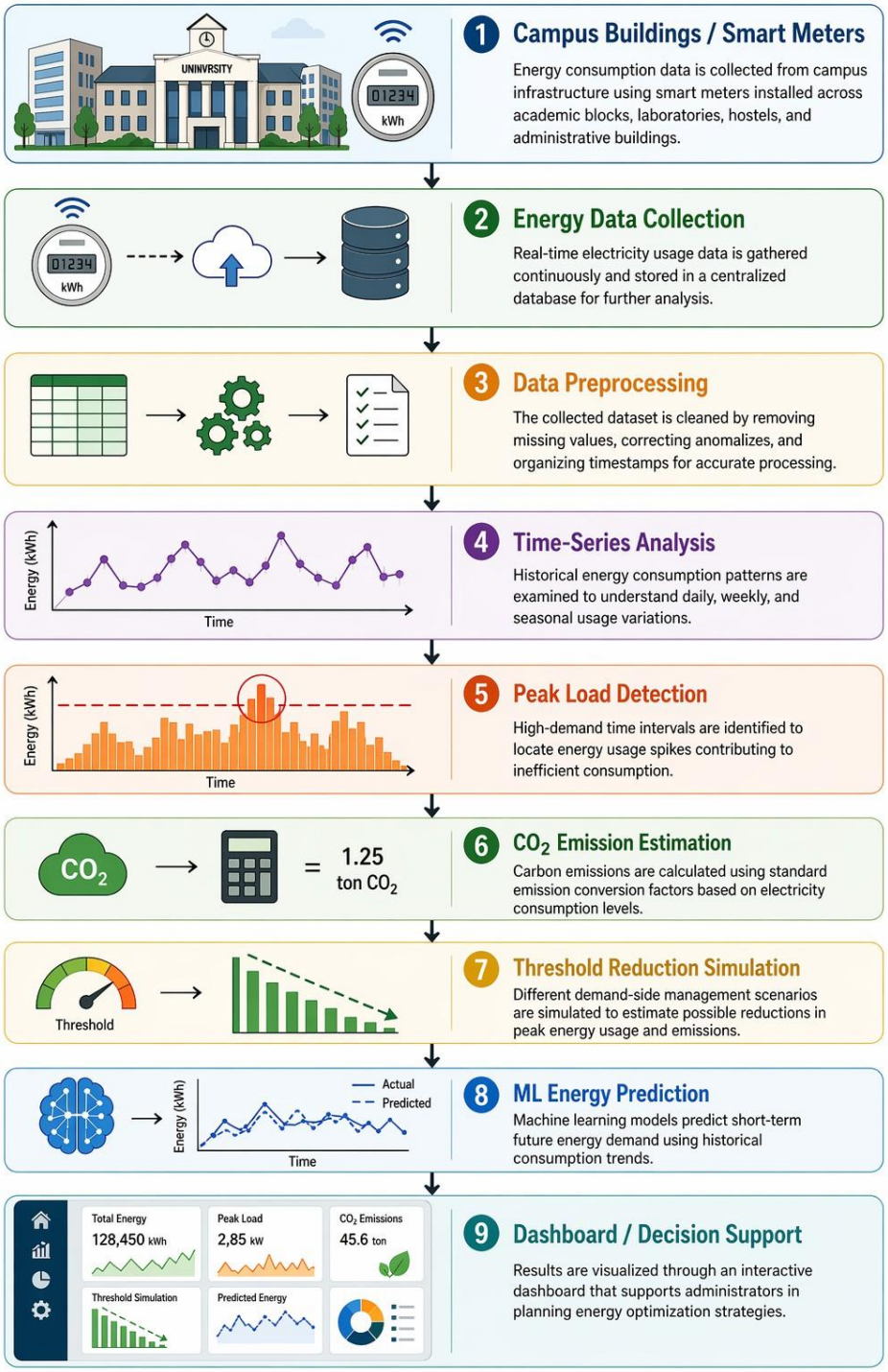
informed demand control strategies can contribute significantly to reducing unnecessary electricity consumption in institutional environments. The model highlights the effectiveness of integrating predictive analytics with building-level monitoring systems for identifying inefficiencies that are otherwise difficult to detect through conventional reporting mechanisms. Furthermore, the framework illustrates how intelligent scheduling and adaptive load management policies can improve overall campus energy performance while supporting long-term environmental sustainability goals.

Another important contribution of this work lies in its scalability and adaptability for deployment across diverse infrastructure environments. Since the framework relies on modular analytics rather than specialized hardware installations, it can be extended to multiple campus blocks, administrative buildings, and smart-city-scale institutional clusters with minimal modification. The proposed methodology therefore offers a cost-effective and practical solution for organizations seeking to implement carbon monitoring strategies aligned with national and global sustainability initiatives.

In addition to supporting real-time monitoring objectives, the framework also creates opportunities for integrating renewable energy forecasting models, machine learning-based load prediction techniques, and automated control mechanisms in future smart campus implementations. Such extensions can further strengthen the role of intelligent monitoring systems in achieving energy-efficient operations and reducing environmental impact across large institutional infrastructures.

Overall, the proposed carbon footprint monitoring system demonstrates the potential of IoT-enabled analytical approaches in transforming traditional campus energy management practices into intelligent, data-driven sustainability frameworks capable of supporting environmentally responsible decision-making in modern smart infrastructure ecosystems.

GRAPHICAL ABSTRACT



ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
DSM	Demand Side Management
HVAC	Heating, Ventilation and Air Conditioning
CO_2	Carbon Dioxide
BMS	Building Management System
EMS	Energy Management System
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory Network
kWh	Kilowatt-hour
GHG	Greenhouse Gas
API	Application Programming Interface
CSV	Comma Separated Values
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
ICT	Information and Communication Technology
PV	Photovoltaic
NLP	Natural Language Processing

SYMBOLS

Symbol	Description	Unit
E	Energy consumption	kWh
C	Carbon emission	$kgCO_2$
EF	Emission factor	$kgCO_2/kWh$
P(t)	Power consumption at time t	kW
T	Time duration	hours
O	Occupancy level	persons
$E_{baseline}$	Baseline energy consumption	kWh
$E_{optimized}$	Optimized energy consumption	kWh
ΔE	Energy savings	kWh
C_{total}	Total carbon emission	$kgCO_2$
α	Learning rate parameter	–
L(t)	Loss function value	–
W	Model weight parameter	–
IoU	Intersection over Union (evaluation metric)	–
MAE	Mean Absolute Error	–
RMSE	Root Mean Square Error	–

CHAPTER 1
INTRODUCTION

1.1 Client / Need Identification and Relevant Contemporary Issues

University campuses today operate almost like small urban systems. They include classrooms, laboratories, auditoriums, libraries, hostels, administrative blocks, computer centres, server rooms, and other shared facilities. All these spaces consume electricity throughout the day, and in some cases even during non-teaching hours. This continuous use of electricity increases the overall carbon footprint of the institution and creates a need for smarter energy monitoring practices.

In most campuses, electricity consumption is still monitored mainly through monthly bills or broad meter readings. Such methods show how much energy has been consumed, but they do not clearly explain when, where, and why higher consumption is occurring. For example, a laboratory may continue to consume power after working hours, or HVAC systems may operate even when classrooms are only partially occupied. These small inefficiencies, when repeated across many buildings, can produce a considerable increase in energy use and carbon emissions.

The client or end-user need in this project can therefore be understood from the perspective of a university administration or campus facility management team. Such users require a system that can help them monitor energy consumption, identify peak load periods, estimate carbon emissions, and support better decision-making for reducing avoidable energy usage. The system should not merely record electricity data but should convert it into useful insights.

The proposed project, titled **Carbon Footprint Monitoring Framework for Smart Campuses Using IoT and Occupancy Analytics**, addresses this need by developing a data-driven analytical framework for campus energy monitoring. The framework uses historical time-series electricity consumption data to identify demand patterns and estimate carbon emissions. It also applies threshold-based reduction scenarios to evaluate how much carbon reduction may be achieved during peak consumption periods.

The project is relevant to contemporary issues such as climate change, sustainable infrastructure, smart campuses, and energy-efficient building operation. With the increasing use of IoT-enabled smart meters, occupancy sensors, and energy analytics platforms, educational institutions can move from static energy reporting to intelligent monitoring. This approach supports sustainability goals without requiring immediate large-scale hardware changes.

1.2 Identification of Problem

Although many educational institutions aim to become sustainable or green campuses, their energy management systems often lack detailed analytical capability. The following problems were identified in the context of campus-level carbon footprint monitoring.

1.2.1 Limited Energy Usage Visibility

Traditional energy monitoring methods usually provide only total electricity consumption values. These values do not clearly show building-level or hourly consumption patterns. As a result, it becomes difficult for administrators to identify which periods or facilities contribute most to energy wastage.

1.2.2 Lack of Occupancy-Based Analysis

Energy usage in campus buildings depends strongly on occupancy. A classroom, laboratory, or auditorium may consume electricity differently depending on whether it is fully occupied, partially occupied, or vacant. However, many existing systems do not consider occupancy conditions while analyzing electricity consumption. This creates difficulty in identifying avoidable loads.

1.2.3 Difficulty in Identifying Peak Load Periods

Peak load periods are important because they often contribute disproportionately to carbon emissions and operational cost. Without time-series analysis, institutions may not clearly understand when these high-demand periods occur. This limits the effectiveness of demand-side management strategies.

1.2.4 Absence of Carbon Emission Estimation

Most electricity monitoring systems focus only on energy consumption in kWh. They do not always translate this energy usage into equivalent carbon emissions. Without carbon emission estimation, it becomes difficult to measure the environmental impact of campus operations.

1.2.5 High Cost of Full IoT Deployment

A complete IoT-based monitoring system may require smart meters, sensors, communication networks, cloud platforms, and dashboards. For many institutions, this can be expensive at the initial

stage. Therefore, there is a need for a framework that can first analyze historical energy data and simulate reduction strategies before large-scale hardware deployment.

These problems define the scope of the proposed project. The system developed in this work does not claim to replace a complete building automation system. Instead, it provides an analytical and simulation-based approach that can support future IoT integration and sustainability planning.

1.3 Identification of Tasks

The following tasks were identified to design, implement, and validate the proposed carbon footprint monitoring framework.

Table 1.1: Task Identification and Allocation

Task No.	Task Description	Owner	Duration
T1	Literature review on smart campus energy monitoring and IoT frameworks	All Members	2 Weeks
T2	Dataset collection and preprocessing of campus electricity consumption records	Misha Kumari, Manasvi Sharma	2 Weeks
T3	Time-series structuring and hourly peak-load analysis	Simardeep Singh Puri	1 Week
T4	Carbon emission estimation using emission factor methodology	Garvit Madaan	1 Week
T5	Threshold-based reduction simulation for 5%, 10%, 15%, and 20% scenarios	All Members	2 Weeks
T6	Machine learning prediction using Random Forest Regression	Simardeep Singh Puri	2 Weeks
T7	System architecture design using layered IoT analytics framework	Manasvi Sharma	1 Week
T8	Visualization of carbon emission trends and reduction results	Misha Kumari	1 Week
T9	Result validation and comparative scenario analysis	All Members	2 Weeks

1.4 Timeline

The project was executed over a twelve-week period following a structured plan. The initial phase focused on literature review and understanding existing work related to IoT-based energy monitoring, smart campus systems, demand-side management, and carbon emission estimation.

The second phase involved dataset collection and preprocessing. During this stage, the electricity consumption data was cleaned, formatted, and arranged into suitable time-series form. This step was necessary because accurate energy and carbon analysis depends on reliable input data.

The third phase focused on hourly consumption analysis and peak-load identification. After identifying the periods of high demand, threshold-based reduction simulations were performed. These simulations helped estimate possible emission reductions under different load reduction percentages.

The fourth phase included machine learning-based energy prediction, visualization of results, and validation of the proposed analytical framework. The final phase focused on preparing the project report, presentation, and supporting documentation.

The Gantt chart for the project timeline is prepared separately using Excel and inserted in this section as Figure 1.1.

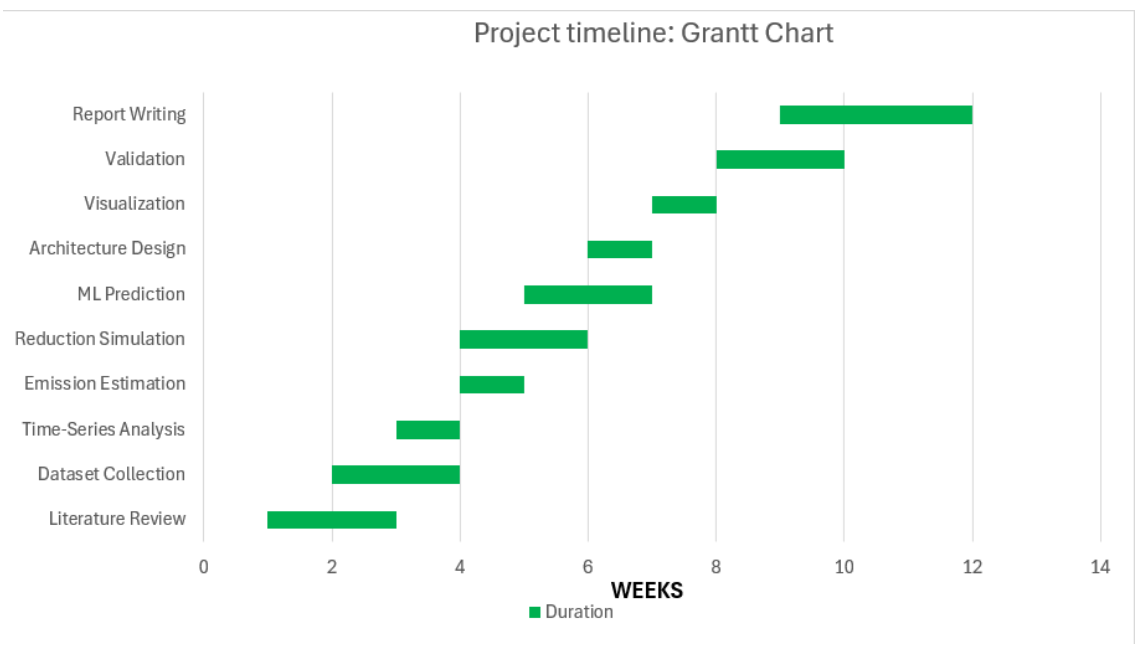


Figure 1.1: Gantt Chart Showing Project Timeline Over Twelve Weeks

1.5 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2 – Literature Survey:** Presents a review of existing research related to IoT-based smart building systems, energy monitoring, occupancy-based control, carbon emission estimation, and demand-side management. It also links the literature review with the problem definition, goals, and objectives of the project.

- **Chapter 3 – Design Flow / Process:** Describes the concept generation, evaluation of features, design constraints, alternative design approaches, best design selection, and implementation plan of the proposed carbon footprint monitoring framework.

- **Chapter 4 – Results Analysis and Validation:** Presents the implementation of the analytical model, data processing workflow, carbon emission estimation, threshold-based reduction results, machine learning prediction output, and validation of the proposed system.

- **Chapter 5 – Conclusion and Future Work:** Summarizes the outcomes of the project, discusses deviations from expected results, and presents possible future enhancements such as real-time IoT integration, renewable energy forecasting, and smart dashboard deployment.

- **References:** Lists all research papers, reports, and technical sources referred to during the development of the project.

- **Appendix / User Manual:** Provides supplementary details, implementation support material, and step-by-step instructions required to understand and run the proposed system.

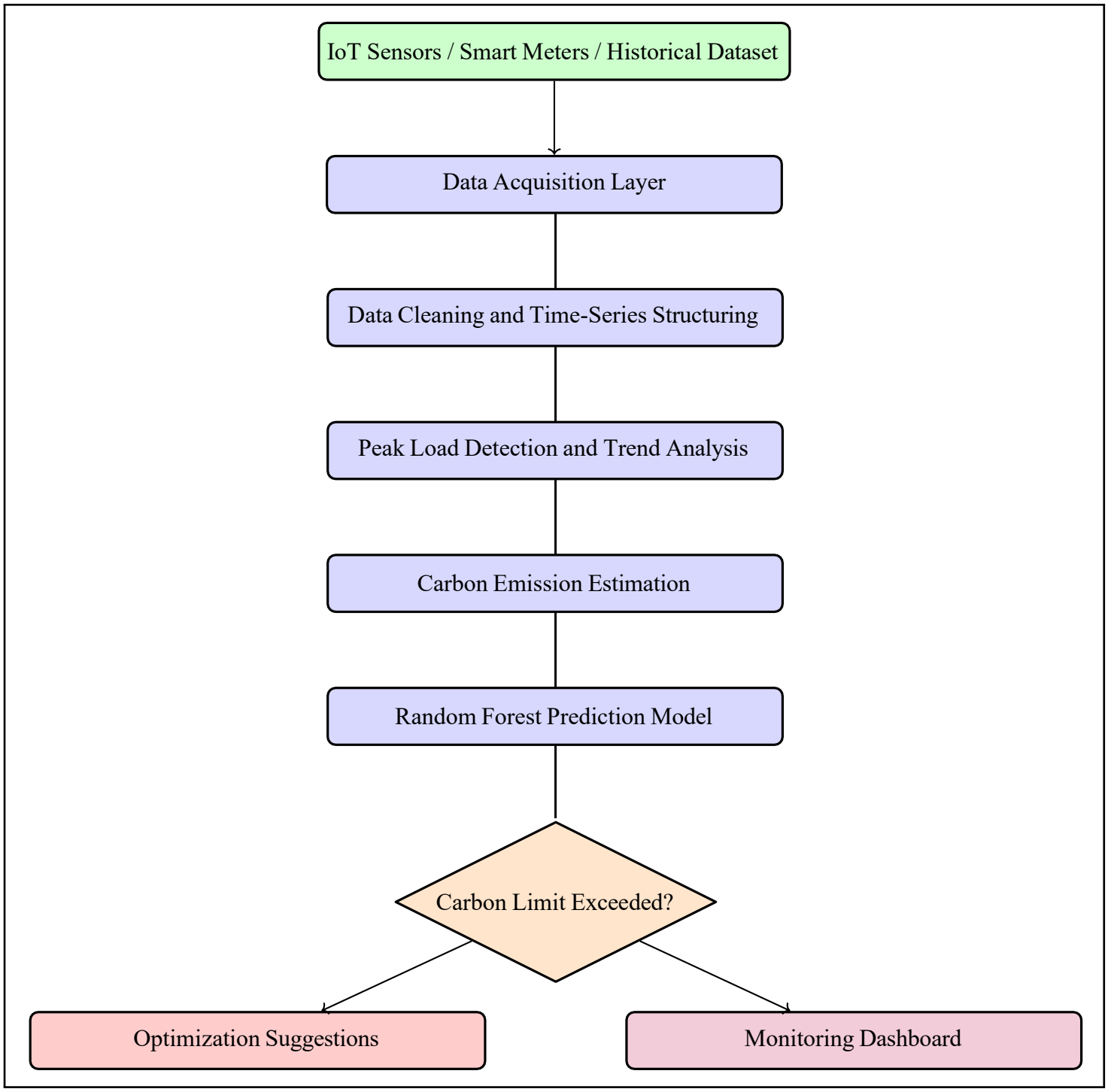


Figure 1.2: Overall Architecture of the IoT-Based Carbon Footprint Monitoring Framework

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction

This chapter presents the literature survey related to IoT-based energy monitoring, smart campus energy management, occupancy-based control, demand-side management, carbon emission estimation, and machine learning-based energy prediction. The purpose of this review is to understand existing research approaches and identify the gap addressed by the proposed project.

The project focuses on a carbon footprint monitoring framework for smart campuses using IoT and occupancy analytics. The research paper developed for this project uses historical time-series electricity consumption data, threshold-based demand-side management, IPCC-based carbon emission estimation, and Random Forest Regression for short-term energy prediction. These elements are also reflected in the literature discussed in this chapter. :contentReference[oaicite:0]index=0

2.2 Timeline of the Reported Problem

The problem of energy monitoring and carbon emission reduction has developed gradually with the growth of smart buildings, IoT systems, and sustainability-focused infrastructure. Earlier research mainly focused on electricity monitoring and occupancy-based lighting control. Later studies introduced smart meters, real-time dashboards, IoT-enabled sensing, and machine learning-based forecasting.

Occupancy-based control was one of the early approaches used to reduce energy wastage in buildings. Neida, Maniccia, and Tweed studied the energy and cost-saving potential of occupancy sensors in commercial lighting systems. Their work showed that energy usage can be reduced when lighting systems respond to actual space utilization instead of remaining active continuously.

With the development of IoT, researchers began using sensor-based systems for real-time energy monitoring. IoT-enabled smart building platforms allowed energy consumption data to be collected continuously from meters, sensors, and building automation systems. These systems improved the ability to detect abnormal consumption and identify energy-saving opportunities.

Recent studies have shifted toward smart campus and net-zero building frameworks. These works combine IoT monitoring, renewable energy integration, predictive analytics, and decision-support systems. In the context of university campuses, such frameworks are important because

campuses operate like small urban environments with classrooms, laboratories, hostels, libraries, and administrative buildings consuming electricity throughout the day.

2.3 IoT-Based Smart Building Energy Monitoring

IoT-based energy monitoring systems use smart meters, sensors, communication networks, and data analytics platforms to collect and analyze building energy consumption. These systems are useful because they provide more detailed information than traditional electricity bills or manual readings.

Sivasankari and Rathika discussed IoT-driven building automation systems and explained how smart lighting, HVAC control, occupancy-based automation, and predictive analytics can improve energy efficiency and sustainability. Their work highlights that IoT systems can reduce unnecessary electricity usage by adjusting building systems according to real-time demand.

Ahmed et al. explored IoT-based smart building systems for improving energy efficiency. Their work shows that sensor-based monitoring can help building managers understand consumption patterns and identify avoidable energy wastage. This supports the idea that smart campuses can benefit from continuous monitoring and data-driven energy decisions.

In the proposed project, IoT is considered as a future integration layer. The current implementation mainly uses historical electricity consumption data, but the framework is designed so that smart meters and occupancy sensors can be connected later for real-time monitoring.

2.4 Occupancy-Based Energy Optimization

Occupancy plays an important role in building energy consumption. Spaces such as classrooms, laboratories, seminar halls, libraries, and offices do not have uniform usage throughout the day. If energy systems operate at full capacity even when occupancy is low, significant energy wastage occurs.

Earlier studies on occupancy-based control systems showed that sensors can reduce lighting and HVAC energy usage by adjusting operation according to actual space utilization. For example, lights can be dimmed or switched off in empty classrooms, and HVAC systems can be reduced during low occupancy hours.

In the proposed framework, occupancy analytics is considered conceptually through peak-load and threshold-based reduction logic. Instead of installing physical occupancy sensors during the current stage, the system estimates the effect of controlled energy reduction during peak demand periods. This makes the project feasible while still supporting future integration with real-time occupancy sensors.

2.5 Smart Campus Energy Management

Smart campuses are large institutional environments where energy consumption is distributed across multiple facilities. Universities include academic blocks, hostels, laboratories, data centers, cafeterias, libraries, and administrative areas. Because of this diversity, energy consumption patterns are complex and vary according to time, occupancy, and activity type.

Franco et al. studied energy savings in university buildings and discussed the role of smart monitoring and IoT technologies. Their work reported that real-time monitoring can support energy reduction in institutional buildings. Such findings are important because university campuses are suitable environments for testing smart energy management systems.

Garcia-Monge Rabanos et al. investigated whether IoT monitoring can improve building energy efficiency through a smart campus case study. Their research supports the idea that sensor-based monitoring can help institutions improve sustainability performance through better data visibility and control.

The proposed project builds on these ideas by developing a framework that can analyze campus energy consumption, estimate carbon emissions, and simulate possible reductions using demand-side management strategies.

2.6 Demand-Side Management and Peak Load Reduction

Demand-side management refers to strategies used to control or reduce electricity demand from the consumer side. Instead of only increasing power generation, DSM focuses on reducing unnecessary usage, shifting loads, and managing peak demand.

Palensky and Dietrich discussed demand response, intelligent energy systems, and smart loads. Their work explains how demand-side strategies can reduce pressure on power systems during high-consumption periods. This is directly relevant to the proposed project because peak-load intervals are the main target for carbon reduction simulation.

In the proposed system, peak-load intervals are identified from hourly electricity consumption data. Reduction thresholds of 5%, 10%, 15%, and 20% are then applied to these high-demand intervals. This helps estimate how much carbon emission reduction may be achieved through controlled demand reduction.

2.7 Carbon Emission Estimation Methods

Carbon emission estimation is necessary to convert electricity consumption into environmental impact. Electricity usage alone only shows energy demand, while carbon emission estimation

shows the amount of CO_2 associated with that demand.

The Intergovernmental Panel on Climate Change provides widely accepted guidelines for greenhouse gas inventory preparation. In the proposed project, electricity consumption values are converted into CO_2 emissions using the emission factor method.

The basic calculation used is:

$$CO_2 = E \times EF$$

where E represents electricity consumption in kWh and EF represents the emission factor in $kgCO_2/kWh$.

This method is simple, standardized, and suitable for campus-level carbon footprint estimation. It allows the system to compare baseline emissions with optimized emissions after threshold-based reduction.

2.8 Machine Learning for Energy Prediction

Machine learning techniques are widely used for energy demand forecasting in smart buildings and campus environments. Energy consumption patterns depend on several factors such as time of day, occupancy levels, seasonal variation, academic schedules, equipment usage, and HVAC operation. Unlike traditional statistical methods, machine learning models can capture nonlinear relationships within historical datasets and improve short-term prediction accuracy.

Deb et al. reviewed time-series forecasting techniques for building energy consumption and highlighted the importance of predictive models in supporting intelligent energy management and proactive demand control strategies.

Several studies have applied algorithms such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), Decision Trees, and Random Forest models for electricity demand prediction. Among these, Random Forest Regression has shown strong performance due to its ability to reduce overfitting and handle multiple input variables effectively.

In the proposed framework, Random Forest Regression is used for short-term energy demand prediction using temporal features such as hour of the day and sequential consumption patterns. This enables identification of recurring peak-demand intervals and supports demand-side management strategies.

Predictive analytics also strengthens threshold-based carbon reduction simulations by estimating future consumption trends under different reduction scenarios. As a result, the proposed system moves beyond static monitoring toward predictive energy intelligence and remains suitable for future IoT-enabled smart campus deployment.

2.9 Comparison of Existing Research

Table 2.1: Comparison of Existing Research Approaches

Author / Study	Focus Area	Method Used	Limitation
Sivasankari and Rathika	IoT-based building automation	Smart lighting, HVAC control, occupancy analytics	Mostly review-based, limited campus-level simulation
Ahmed et al.	Smart building energy efficiency	IoT-based monitoring and sensor integration	Focused on monitoring rather than threshold simulation
Fakhabi et al.	IoT in green buildings	IoT and renewable energy integration	Requires infrastructure-level implementation
Franco et al.	University building energy savings	Real-time monitoring in campus buildings	Results depend on existing building performance
Neida et al.	Occupancy-based lighting control	Occupancy sensors for energy saving	Limited to lighting systems
Palensky and Dietrich	Demand-side management	Demand response and smart loads	General DSM framework, not campus-specific
Deb et al.	Building energy forecasting	Time-series forecasting review	Does not directly estimate carbon reduction
Proposed Project	Smart campus carbon monitoring	Time-series analysis, threshold simulation, IPCC emission estimation, Random Forest prediction	Real-time IoT deployment kept as future scope

2.10 Summary Linking Literature Review with the Project

The literature review shows that IoT-based monitoring, occupancy-based control, demand-side management, and machine learning prediction are all important for improving building energy efficiency. Existing studies confirm that smart monitoring can reduce electricity wastage and support sustainable building operation.

However, many existing works focus either on hardware-based monitoring or on general energy efficiency without combining carbon emission estimation, threshold-based reduction simulation, and machine learning prediction in a single campus-oriented framework. This creates a research

gap.

The proposed project addresses this gap by developing a data-driven framework that uses historical campus energy data to identify peak-load intervals, estimate carbon emissions, simulate reduction scenarios, and predict short-term energy demand. The approach is practical because it does not require immediate hardware installation, but it remains scalable for future IoT deployment.

2.11 Problem Definition

Large university campuses consume significant electricity because of classrooms, laboratories, hostels, libraries, offices, HVAC systems, data centers, and other facilities. Traditional monitoring methods usually provide aggregate electricity usage but do not provide detailed insights into hourly consumption, peak-load behavior, occupancy-related wastage, or carbon emission impact.

Therefore, the problem addressed in this project is the lack of a scalable and data-driven framework for monitoring campus electricity consumption, estimating associated carbon emissions, identifying peak-load intervals, and simulating energy reduction strategies that can support smart campus sustainability planning.

2.12 Goals and Objectives

The main goal of this project is to develop a carbon footprint monitoring framework for smart campuses using IoT-compatible architecture and occupancy-aware analytics.

The objectives of the project are:

- To analyze historical campus electricity consumption data using time-series methods.
- To preprocess and structure electricity data into hourly intervals.
- To identify peak-load periods responsible for high energy demand.
- To estimate carbon emissions using emission factor-based calculation.
- To simulate carbon reduction under 5%, 10%, 15%, and 20% threshold reduction scenarios.
- To apply Random Forest Regression for short-term energy demand prediction.
- To design a scalable framework that can be integrated with IoT-enabled smart meters in the future.
- To support campus administrators in making data-driven sustainability decisions.

2.13 Research Gap

Based on the reviewed literature, several important research gaps have been identified in the area of smart campus energy monitoring and carbon footprint estimation.

- Many existing smart building energy management systems primarily focus on electricity consumption monitoring without performing detailed carbon emission estimation using standardized emission factor methodologies. As a result, the environmental impact of energy usage is not fully quantified.
- A large number of IoT-based energy monitoring frameworks rely heavily on real-time sensor deployment, smart meters, and building automation infrastructure. Such implementations are often expensive and difficult to deploy in educational institutions with limited technical resources.
- Several research studies treat energy consumption analysis and machine learning-based prediction as independent problems. Integrated frameworks that combine monitoring, forecasting, and emission reduction simulation within a single architecture are still limited.
- Existing demand-side management approaches often focus on load shifting or peak shaving without estimating the corresponding reduction in carbon emissions. This creates a gap between operational optimization and sustainability assessment.
- Most smart campus energy frameworks emphasize real-time control strategies but do not explore simulation-based reduction scenarios such as threshold-driven peak-load optimization using historical datasets.
- Limited research has been conducted on campus-scale analytical frameworks that integrate time-series energy analysis with emission estimation and predictive modeling while remaining compatible with future IoT deployment.
- Many current sustainability monitoring systems depend on complex infrastructure integration and cloud-based automation platforms, which may not be practical for early-stage institutional adoption or academic implementation.
- There is insufficient work on scalable analytical models that allow institutions to evaluate multiple emission reduction scenarios (such as 5%, 10%, 15%, and 20% demand reduction) before implementing physical energy-saving interventions.

- Existing studies rarely provide modular layered architectures that support gradual transition from dataset-driven simulation models to fully IoT-enabled smart campus monitoring environments.

The proposed project addresses these research gaps by developing a simulation-based and scalable carbon footprint monitoring framework that integrates time-series energy consumption analysis, emission factor-based carbon estimation, threshold-based reduction modeling, and machine learning-based prediction. The framework is designed to operate using historical datasets in its current form while remaining extensible for future integration with IoT-enabled smart meters and occupancy-aware sensing systems.

2.14 Conclusion

This chapter reviewed existing studies related to IoT-enabled energy monitoring, occupancy analytics, smart campus systems, demand-side management, carbon emission estimation, and machine learning-based prediction. The review helped identify the need for a framework that combines historical energy analysis, carbon estimation, threshold-based optimization, and future IoT compatibility. The next chapter presents the design flow and process followed in developing the proposed framework, including feature selection, design constraints, alternative designs, final design selection, and implementation methodology.

CHAPTER 3
DESIGN FLOW / PROCESS

3.1 Evaluation and Selection of Specifications / Features

This chapter presents the systematic design process followed for developing the proposed carbon footprint monitoring framework for smart campuses. The framework is based on the analysis of campus electricity consumption data, estimation of carbon emissions, identification of peak-load intervals, threshold-based reduction simulation, and machine learning-based energy demand prediction.

The proposed system is analytical and simulation-based. It does not depend on immediate hardware deployment, although it is designed in such a way that it can be integrated with IoT-enabled smart meters and occupancy sensors in future smart campus environments. The major features were selected after considering feasibility, performance impact, complexity, scalability, and relevance to sustainability monitoring.

3.1.1 Time-Series Energy Consumption Analysis

Time-series energy consumption analysis is one of the most important features of the proposed framework. Since campus electricity usage varies throughout the day depending on classroom schedules, laboratory usage, hostel activity, office hours, and HVAC operation, the energy data must be arranged in chronological form. This allows the system to observe hourly demand patterns and identify repeated peak-load intervals.

The energy dataset used in the project contains time-stamped electricity consumption values. These values are processed and aggregated into hourly intervals so that the system can clearly identify when electricity demand is higher than normal. This feature is essential because peak demand intervals contribute significantly to carbon emissions.

3.1.2 Data Preprocessing

Data preprocessing was selected as a necessary feature because raw energy datasets may contain missing values, inconsistent timestamps, duplicate records, or irregular time intervals. If such errors are not handled properly, the final emission analysis may become unreliable.

The preprocessing stage includes timestamp standardization, missing value handling, hourly aggregation, and data cleaning. This ensures that the input data used for peak-load detection and emission estimation is consistent and suitable for analysis.

3.1.3 Peak Load Detection

Peak load detection helps identify the periods during which campus electricity consumption is unusually high. These periods are important because reducing energy demand during peak hours can produce a noticeable reduction in carbon emissions.

In the proposed framework, peak-load intervals are detected using threshold-based logic. Consumption values above the selected threshold are treated as high-demand intervals. These intervals are then used for carbon reduction simulation.

3.1.4 Carbon Emission Estimation

Carbon emission estimation converts electricity consumption into equivalent carbon dioxide emissions. The proposed system uses an emission factor-based method in which energy consumption in kWh is multiplied by the emission factor to estimate CO_2 emissions.

This feature allows the system to move beyond simple electricity monitoring and provide environmental impact analysis. It is useful for campus administrators who want to evaluate energy usage from a sustainability perspective.

3.1.5 Threshold-Based Reduction Simulation

Threshold-based reduction simulation was selected to estimate possible emission savings without requiring immediate physical infrastructure changes. In this approach, different reduction levels such as 5%, 10%, 15%, and 20% are applied to peak-load intervals.

The revised energy consumption and carbon emissions are then recalculated. This helps estimate how much carbon emission reduction can be achieved if campus energy demand is reduced during high-consumption periods.

3.1.6 Machine Learning-Based Energy Prediction

A machine learning prediction module was included to support short-term energy demand forecasting. Random Forest Regression was selected because it is suitable for continuous numerical prediction and can handle time-based input features.

The model uses historical energy consumption data along with temporal features such as hour of the day and day information. The predicted energy demand can help identify possible future peak-load periods and support proactive demand-side management.

3.1.7 Decision Support and Visualization

The final feature selected was decision support through result visualization. Graphs, comparison tables, and emission reduction outputs help users understand the impact of different energy-saving scenarios. This makes the framework useful not only for analysis but also for planning future smart campus energy policies.

Table 3.1: Feature Evaluation Matrix

Feature	Feasibility	Impact	Complexity	Selected
Time-Series Analysis	High	High	Low	Yes
Data Preprocessing	High	High	Low	Yes
Peak Load Detection	High	High	Medium	Yes
Carbon Emission Estimation	High	High	Low	Yes
Threshold-Based Simulation	High	High	Medium	Yes
Random Forest Prediction	Medium	High	Medium	Yes
Real-Time IoT Integration	Medium	High	High	Future Scope
Dashboard Visualization	High	Medium	Low	Yes
Renewable Energy Integration	Medium	High	High	Future Scope

3.2 Design Constraints

The design of the proposed carbon footprint monitoring framework was influenced by several constraints. These constraints were considered to make the framework practical, scalable, cost-effective, and suitable for academic implementation.

3.2.1 Regulatory Constraints

The carbon emission estimation process must follow accepted environmental accounting practices. In this project, the emission factor-based approach was selected because it aligns with standard carbon accounting methods. The framework is designed to estimate electricity-related CO_2 emissions using energy consumption and emission factor values.

3.2.2 Economic Constraints

A complete IoT-based smart campus system requires smart meters, occupancy sensors, cloud infrastructure, communication modules, and dashboard deployment. Such hardware installation can be expensive for an academic project. Therefore, the current framework uses historical energy consumption data and simulation-based analysis. This reduces cost while still allowing meaningful sustainability assessment.

3.2.3 Technical Constraints

The major technical limitation was the unavailability of real-time campus sensor data. Because of this, the system was designed using historical time-series electricity consumption records. Another technical constraint was the need to keep the model simple enough for academic implementation while still including useful analytics such as peak-load detection and machine learning prediction.

3.2.4 Environmental Constraints

Energy consumption in campus buildings is affected by occupancy levels, seasonal variation, classroom schedules, laboratory usage, HVAC loads, and working hours. The framework must therefore account for variation in electricity demand instead of assuming constant consumption throughout the day.

3.2.5 Health and Safety Constraints

Since the present implementation is based on data analysis and simulation, it does not create direct health or safety risks. However, if implemented in a real campus with IoT devices, the system must ensure safe electrical installation of smart meters and sensors.

3.2.6 Ethical Constraints

The framework does not use personal data, facial images, student tracking, or individual occupancy records. It relies only on aggregated electricity consumption patterns. This makes the system ethically safer compared to monitoring systems that collect personal movement or identity-related data.

3.2.7 Professional Constraints

The project must follow proper academic documentation, modular design, and reproducible methodology. The implementation should be clear enough for evaluation, verification, and future extension.

Table 3.2: Design Constraints Summary

Constraint Type	Specific Constraint	Impact on Design
Regulatory	Use of standard emission estimation method	Emission factor-based CO_2 calculation selected
Economic	No large-scale hardware deployment	Historical dataset and simulation-based framework used
Technical	Real-time smart meter data unavailable	Time-series historical data used as input
Environmental	Variation in building usage and occupancy	Peak-load and hourly analysis included
Health and Safety	Future sensor deployment must be electrically safe	Hardware integration kept as future scope
Ethical	Avoid personal or sensitive data collection	Aggregated energy data used only
Professional	Report must be reproducible and well documented	Modular workflow and clear methodology adopted

3.3 Analysis and Feature Finalization Subject to Constraints

After evaluating the design constraints, the final feature set was selected. Some features were included in the current implementation, while others were kept as future extensions. Since the project is mainly a data-driven carbon monitoring framework, the selected features focus on reliable analysis, emission estimation, and simulation rather than full hardware deployment.

Table 3.3: Feature Finalization after Constraint Analysis

Feature	**Final Decision**	**Reason**
Historical Energy Dataset Analysis	Selected	Available, practical, and suitable for simulation-based study
Real-Time IoT Smart Meter Integration	Future Scope	Requires physical sensor installation and campus-level hardware access
Data Preprocessing	Selected	Required for reliable time-series analysis
Peak Load Detection	Selected	Essential for demand-side management and carbon reduction simulation
Carbon Emission Estimation	Selected	Core objective of the project
Threshold-Based Reduction Simulation	Selected	Allows comparison of 5%, 10%, 15%, and 20% reduction scenarios
Random Forest Prediction	Selected	Supports short-term energy demand forecasting
Renewable Energy Integration	Future Scope	Requires solar generation or renewable source data
Automated Control System	Deferred	Requires connection with building automation systems
Dashboard Visualization	Selected	Helps interpret emission trends and reduction results

These decisions helped keep the project practical and aligned with the available resources. The final framework prioritizes analytical reliability, carbon reduction estimation, and future scalability rather than overpromising a complete physical IoT deployment.

3.4 Design Flow – Alternative Designs

Two possible design alternatives were considered for the proposed carbon footprint monitoring framework. Both alternatives aim to estimate campus carbon emissions, but they differ in terms of complexity, prediction capability, and scalability.

3.4.1 Design Alternative 1 – Statistical Energy Analysis Framework

The first design alternative is based only on statistical analysis of historical electricity consumption data. In this design, the system collects energy data, preprocesses it, identifies hourly consumption trends, detects peak-load intervals, and estimates carbon emissions.

The basic workflow of this design is as follows:

Dataset Collection → Data Preprocessing → Hourly Time-Series Analysis → Peak Load Detection → Carbon Emission Estimation → Reduction Simulation

This design is simple, easy to implement, and computationally efficient. It does not require machine learning algorithms or real-time sensor integration. However, its major limitation is that it only provides retrospective analysis. It can identify past peak-load intervals but cannot predict future energy demand.

Advantages:

- Easy to implement using historical data.
- Low computational complexity.
- No hardware or sensor requirement.
- Suitable for basic carbon emission estimation.

Disadvantages:

- No predictive capability.
- Limited support for proactive decision-making.
- Less scalable for real-time smart campus deployment.

3.4.2 Design Alternative 2 – Hybrid IoT Analytics and Machine Learning Framework

The second design alternative combines time-series analysis, emission estimation, threshold-based simulation, and machine learning prediction. This design is more advanced because it supports both analysis of past data and prediction of future demand.

The basic workflow of this design is as follows:

Energy Dataset / IoT Data → Preprocessing → Time-Series Structuring → Peak Load Detection → Carbon Estimation → Threshold Simulation → Random Forest Prediction → Decision Support

This design was selected for the final project because it provides a stronger foundation for smart campus sustainability monitoring. It can work with historical datasets in the current implementation and can also be extended to real-time IoT data in future deployments.

Advantages:

- Supports both historical analysis and future prediction.
- Can be integrated with IoT-enabled smart meters.
- Helps identify peak-load periods in advance.
- Provides stronger decision support for sustainability planning.

Disadvantages:

- Requires more implementation effort.
- Machine learning prediction depends on data quality.
- Real-time deployment requires additional IoT infrastructure.

Alternative Design Flow 1: Statistical Framework

Dataset Collection → Preprocessing → Time-Series Analysis → Peak Detection → Carbon Estimation → Reduction Simulation

Alternative Design Flow 2: Hybrid IoT-ML Framework

Dataset / Smart Meter Data → Preprocessing → Analytics → Carbon Estimation → ML Prediction → Decision Support

Figure 3.1: Alternative Design Flows for Carbon Footprint Monitoring Framework

3.5 Design Selection

After evaluating both design alternatives, the Hybrid IoT Analytics and Machine Learning Framework was selected as the final design. The reason for selecting this approach is that it provides better scalability, predictive capability, and practical relevance for future smart campus deployment.

Although the statistical framework is simpler, it is limited to analyzing past energy data. The selected hybrid framework can analyze historical data, estimate emissions, simulate carbon reduction, and predict short-term energy demand using Random Forest Regression.

Table 3.4: Design Comparison – Statistical Framework vs Hybrid Framework

Criterion	**Design 1: Statistical Framework**	**Design 2: Hybrid IoT-ML Framework**	**Selected**
Implementation Complexity	Low	Medium	Design 2
Carbon Emission Estimation	Yes	Yes	Both
Peak Load Detection	Yes	Yes	Both
Prediction Capability	No	Yes	Design 2
IoT Integration Support	Limited	High	Design 2
Scalability	Medium	High	Design 2
Cost	Low	Medium	Design 1
Decision Support	Basic	Strong	Design 2
Future Expansion	Limited	High	Design 2

The selected design provides a balanced solution between academic feasibility and practical relevance. It can be implemented using historical datasets at present and can later be extended into a real-time IoT-enabled campus energy management system.

3.6 Implementation Plan / Methodology

The implementation of the proposed framework follows a structured and modular analytical pipeline. Each stage is responsible for processing a specific part of the workflow, beginning from dataset preparation and ending with decision-support outputs. The methodology integrates time-series analysis, emission factor-based carbon estimation, demand-side management simulation, and machine learning prediction within a unified analytical structure. This stepwise approach ensures that the framework remains scalable and suitable for future integration with IoT-enabled monitoring systems.

3.6.1 Step 1 – Dataset Collection

The first step involves collecting historical electricity consumption data representing campus buildings or institutional energy usage. The dataset contains timestamped energy consumption values that allow detailed time-series analysis. These records form the foundation for identifying consumption patterns and evaluating emission reduction scenarios. The dataset structure is selected to resemble real-world smart campus monitoring environments.

3.6.2 Step 2 – Data Preprocessing

The collected dataset is cleaned and formatted before analysis. Timestamp fields are standardized into uniform time intervals, missing or inconsistent entries are handled appropriately, and electricity consumption values are arranged into hourly sequences. This preprocessing stage improves data reliability and ensures compatibility with statistical and machine learning techniques used in later stages.

3.6.3 Step 3 – Time-Series Structuring

After preprocessing, the dataset is converted into a structured time-series format. Hourly energy consumption values are grouped and analyzed to identify daily and periodic demand variations. This helps in understanding peak operational hours of campus facilities and provides the basis for identifying high-consumption intervals requiring optimization.

3.6.4 Step 4 – Peak Load Identification

Peak load intervals are identified by comparing energy consumption values against selected threshold levels derived from the dataset distribution. These high-demand periods are important because

they represent the primary targets for demand-side management strategies. Identifying such intervals enables selective optimization rather than uniform load reduction across all time periods.

3.6.5 Step 5 – Carbon Emission Estimation

Energy consumption values are converted into carbon emissions using the emission factor method based on standard carbon accounting practices. The basic formula used is:

$$CO_2 = E \times EF$$

where E represents energy consumption in kWh and EF represents the emission factor in $kgCO_2/kWh$. This conversion allows the framework to quantify the environmental impact of electricity usage and evaluate sustainability improvements under different reduction scenarios.

3.6.6 Step 6 – Threshold-Based Reduction Simulation

After identifying peak-load intervals, reduction thresholds of 5%, 10%, 15%, and 20% are applied to simulate possible energy savings. These simulated reductions represent realistic demand-side management actions such as optimized scheduling, lighting control, and HVAC adjustment. The revised energy consumption is calculated as:

$$E_{reduced} = E_{peak} \times (1 - T)$$

where T represents the reduction percentage and E_{peak} represents energy consumption during peak intervals. The simulation results help estimate achievable emission reductions without requiring physical infrastructure deployment.

3.6.7 Step 7 – Machine Learning Prediction

Random Forest Regression is applied to forecast short-term energy demand using historical electricity consumption data. Temporal features such as hour of the day, day index, and sequential consumption patterns are used as input variables. The prediction module supports proactive identification of future high-demand periods and strengthens planning capabilities for energy optimization strategies.

3.6.8 Step 8 – Visualization and Decision Support

The final stage involves generating graphical representations, comparison tables, and emission reduction summaries from the analytical results. These visual outputs improve interpretability of energy trends and help administrators evaluate alternative reduction strategies. The visualization layer acts as a decision-support interface that enables informed planning for sustainable smart campus energy management.

Table 3.5: Implementation Workflow of the Proposed Framework

Stage	Description
Dataset Collection	Collection of timestamped electricity consumption data representing campus or institutional buildings. The dataset forms the primary input for time-series structuring, peak-load analysis, and carbon emission estimation.
Data Preprocessing	Cleaning missing values, correcting inconsistencies, formatting timestamps into uniform intervals, and preparing hourly consumption records suitable for analytical processing and machine learning input.
Time-Series Analysis	Structuring electricity usage into hourly sequences to identify daily consumption trends, operational load variations, and recurring peak-demand patterns across campus facilities.
Peak Load Identification	Detection of high-consumption intervals using threshold-based logic derived from dataset distribution. These intervals represent target regions for demand-side management optimization.
Carbon Emission Estimation	Conversion of electricity consumption into equivalent CO_2 emissions using standard emission factor methodology based on IPCC carbon accounting practices.
Threshold Simulation	Application of 5%, 10%, 15%, and 20% peak-load reduction scenarios to simulate achievable energy savings and evaluate their environmental impact under controlled demand-side strategies.
Machine Learning Prediction	Implementation of Random Forest Regression for short-term energy demand forecasting using temporal features such as hour index and sequential consumption patterns.
Decision Support	Visualization of consumption trends, comparison of emission reduction scenarios, and generation of sustainability insights to assist administrators in planning energy optimization strategies.
Framework Integration (Optional IoT Layer)	Provision for future integration with IoT-enabled smart meters and sensor networks to support real-time monitoring, automated control actions, and scalable smart campus deployment.

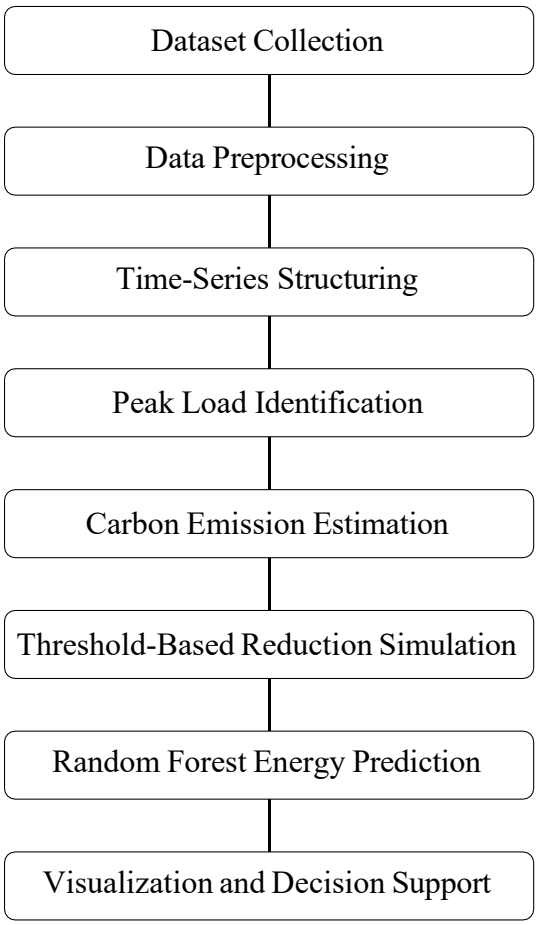


Figure 3.2: Implementation Methodology of the Proposed Carbon Footprint Monitoring Framework

The selected methodology provides a practical and scalable approach for carbon footprint monitoring in campus environments. It enables systematic analysis of energy consumption patterns, estimation of associated carbon emissions, and evaluation of reduction scenarios using simulation-based techniques. At the same time, the framework remains feasible within academic implementation constraints while supporting future integration with IoT-enabled smart meters and occupancy sensing systems.

CHAPTER 4
RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

This chapter presents the implementation details, testing setup, quantitative results, and validation of the proposed carbon footprint monitoring framework. The performance is analysed using building-level carbon emissions, hourly emission trends, occupancy-emission relationship, and threshold-based reduction sensitivity.

The proposed framework was implemented as a data-driven analytical model. It processes historical campus electricity consumption records, estimates carbon emissions, identifies high-emission buildings and peak-load intervals, and evaluates possible emission reduction under different demand-side management scenarios.

4.1.1 Hardware and Software Configuration

Component	Specification
Operating System	Windows / Linux Environment
Programming Language	Python
Data Processing Libraries	Pandas, NumPy
Visualization Tools	Matplotlib, Seaborn
Machine Learning Model	Random Forest Regression
Input Dataset	Historical time-series electricity consumption data
Output Results	Emission trends, building comparison, occupancy correlation, and reduction sensitivity

Table 4.1: Hardware and Software Configuration

4.1.2 Core Implementation – Data Preprocessing Module

The preprocessing module prepares raw electricity consumption data for analysis. Timestamp fields are standardized, missing or inconsistent values are handled, and the dataset is converted

into structured hourly records. This step is essential because the accuracy of carbon emission estimation and peak-load detection depends on clean and consistent input data.

The preprocessing stage includes:

- standardization of timestamp format,
- removal or correction of inconsistent values,
- hourly aggregation of electricity consumption,
- preparation of building-level and occupancy-related records.

4.1.3 Core Implementation – Carbon Emission Estimation Module

Carbon emission estimation was performed by converting electricity consumption values into equivalent CO_2 emissions using the emission factor method. The calculation used in the framework is:

$$CO_2 = E \times EF$$

where E represents electricity consumption in kWh and EF represents the emission factor in $kgCO_2/kWh$.

4.1.4 Core Implementation – Threshold Reduction Module

The threshold reduction module applies controlled reductions to high-consumption intervals. In this project, 5%, 10%, 15%, and 20% threshold scenarios were tested. These values represent possible demand-side management actions such as HVAC adjustment, load scheduling, or controlled lighting operation.

$$E_{reduced} = E_{peak} \times (1 - T)$$

where T is the reduction threshold and E_{peak} is the energy consumption during peak-load intervals.

4.2 Testing and Evaluation Setup

4.2.1 Dataset Description

The framework was evaluated using historical time-series electricity consumption data similar to a campus or institutional building energy dataset. The dataset includes building identifiers, times-tamped energy usage, occupancy estimates, and calculated carbon emission values.

Dataset Attribute	Description	Purpose
Timestamp	Date and time of electricity usage	Hourly trend analysis
Building ID	Unique identifier of each building	Building-wise emission comparison
Energy Consumption	Electricity usage values	CO_2 emission estimation
Occupancy Estimate	Approximate utilization level	Occupancy-emission analysis
Carbon Emission	Calculated CO_2 output	Sustainability evaluation

Table 4.2: Dataset Description

4.2.2 Evaluation Metrics

The effectiveness of the proposed framework was evaluated using the following metrics:

- **Total CO_2 Emissions:** Total calculated carbon output from electricity consumption.
- **Building-Level Emission Contribution:** Identification of buildings with the highest carbon emissions.
- **Occupancy-Emission Relationship:** Analysis of the effect of occupancy estimate on CO_2 output.
- **Percentage Emission Reduction:** Carbon reduction achieved under different threshold scenarios.
- **Absolute Carbon Reduction:** Total avoided CO_2 emissions in tonnes.

4.3 Performance Results

4.3.1 Building-Level Carbon Emission Results

Figure 4.1 presents the top fifteen buildings contributing the highest carbon emissions in the dataset. This comparison helps identify priority buildings for energy optimization.

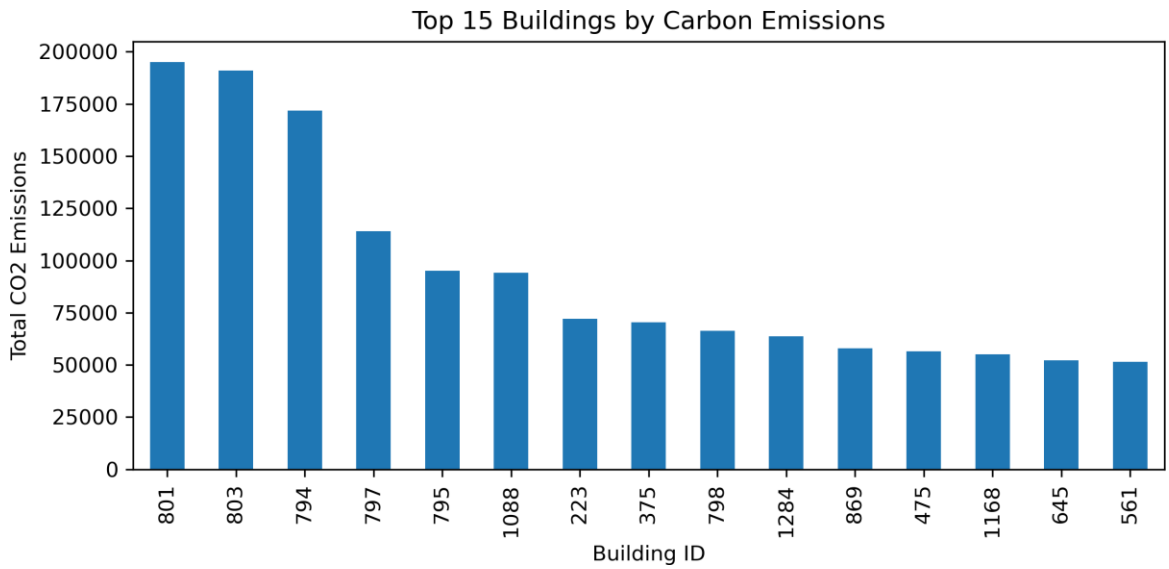


Figure 4.1: Top 15 Buildings by Carbon Emissions

- A small number of buildings contribute a relatively large share of the overall campus carbon footprint, indicating the presence of high-impact emission sources within the infrastructure.
- These buildings typically operate for longer durations and contain higher equipment loads such as centralized HVAC systems, laboratory installations, and continuous electrical usage facilities.
- Higher occupancy levels in academic and laboratory buildings further increase electricity demand, which directly contributes to increased carbon emission intensity.
- The emission distribution pattern observed in Figure 4.1 is non-uniform, confirming the presence of identifiable emission hotspots across campus facilities.
- Targeting high-emission buildings first can produce greater sustainability improvements compared to applying uniform energy optimization strategies across all buildings.
- Building-level emission comparison provides a practical basis for prioritizing demand-side management interventions and phased energy optimization planning.

- The results support the selective deployment of IoT-enabled monitoring systems in high-emission buildings as an effective initial implementation strategy for smart campus infrastructure.
- Such targeted implementation reduces installation cost while maximizing environmental benefits during early-stage sustainability adoption.
- The emission ranking generated by the proposed framework can assist administrators in long-term planning decisions such as retrofit prioritization, equipment replacement scheduling, and adaptive load management policy development.
- These strategies contribute to gradual and sustainable reduction of institutional carbon footprint without affecting normal operational continuity.

4.3.2 Hourly Carbon Emission Trend

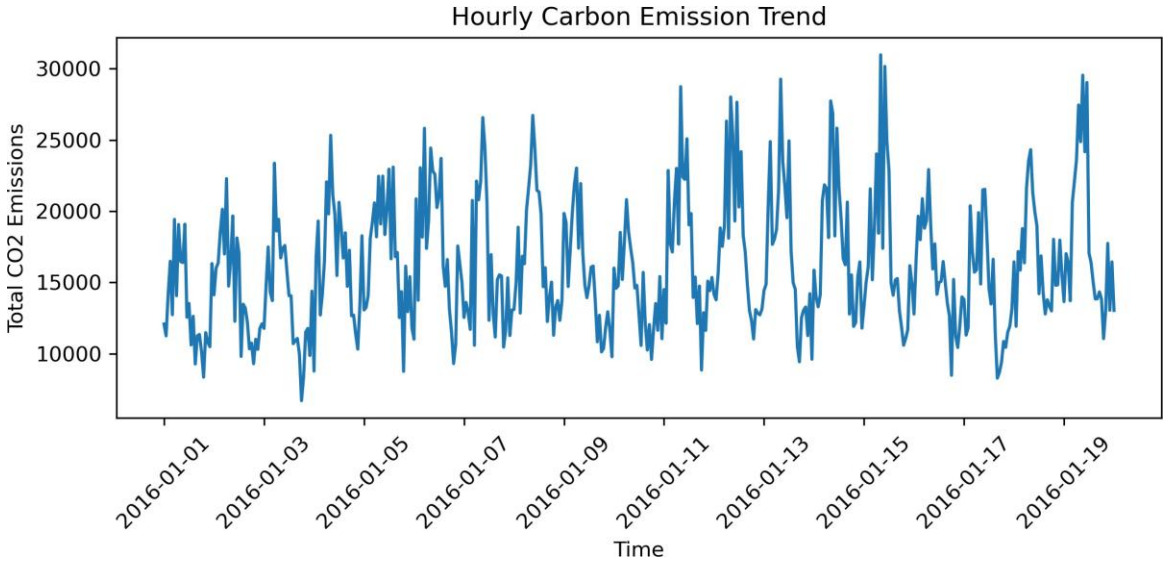


Figure 4.2: Hourly Carbon Emission Trend

The graph shows repeated fluctuations in carbon emissions over time. Higher peaks represent periods of increased electricity demand. These peaks are important because they indicate the time intervals where threshold-based reduction strategies can produce the most effective carbon savings.

4.3.3 Occupancy and Carbon Emission Relationship

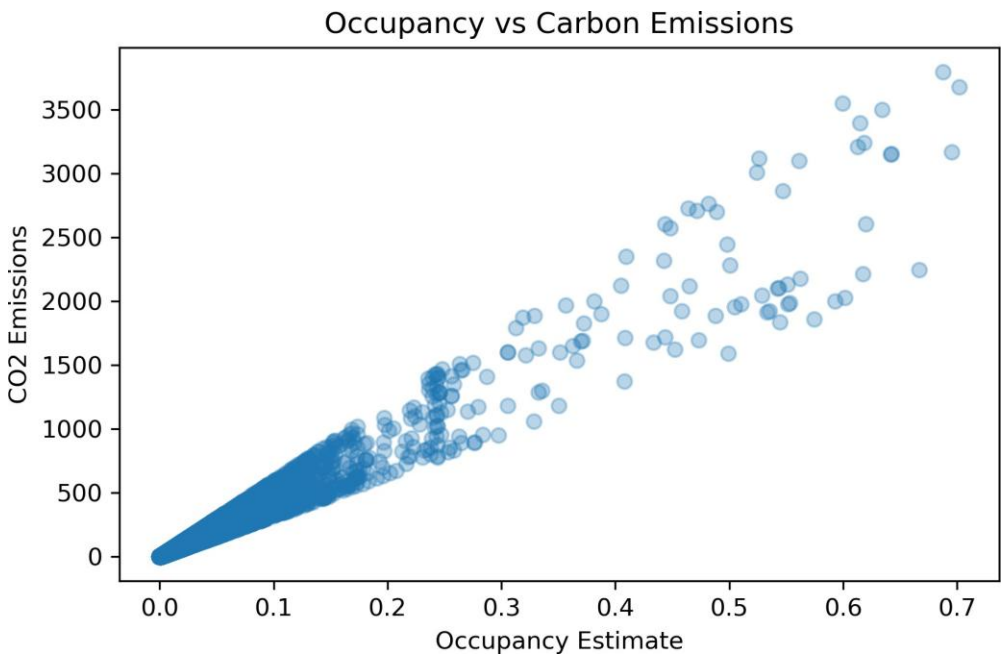


Figure 4.3: Occupancy Estimate versus Carbon Emissions

The scatter plot indicates a positive relationship between occupancy and carbon emissions. As occupancy increases, energy demand also tends to increase due to lighting, cooling, ventilation, and equipment usage. This supports the use of occupancy-aware control strategies in smart campus environments.

4.3.4 Threshold-Based Carbon Reduction Results

The threshold-based simulation evaluated four reduction scenarios: 5%, 10%, 15%, and 20%. These scenarios represent controlled demand-side energy optimization strategies applied during identified peak-load intervals.

The results obtained from the simulation are summarized in Table 4.3.

Threshold Reduction	Emission Reduction	CO_2 Reduction
5%	0.513%	389.45 tonnes
10%	1.543%	1172.11 tonnes
15%	2.873%	2181.85 tonnes
20%	3.882%	2948.52 tonnes

Table 4.3: Carbon Emission Reduction under Different Threshold Scenarios

The simulation results indicate a progressive increase in carbon emission savings as the reduction threshold increases. This confirms that peak-load intervals represent the most effective opportunities for demand-side optimization in campus-scale energy systems. Even a modest reduction in peak consumption can produce measurable environmental benefits without requiring major infrastructure modifications.

The observations from the threshold-based reduction analysis are summarized as follows:

- A 5% reduction in peak-load consumption produced approximately 389.45 tonnes of CO_2 savings, demonstrating that small operational adjustments can contribute meaningfully toward sustainability objectives.
- Increasing the reduction threshold to 10% resulted in carbon savings of approximately 1172.11 tonnes, indicating improved effectiveness of moderate demand-side control strategies.
- A 15% threshold further increased emission reduction to about 2181.85 tonnes, showing that progressive optimization of HVAC scheduling, lighting systems, and equipment utilization can significantly reduce institutional carbon footprint.
- The highest reduction was observed in the 20% threshold scenario, achieving approximately 2948.52 tonnes of CO_2 savings, which represents the upper-bound optimization condition evaluated in this study.
- The results confirm that targeted peak-load management strategies can provide scalable and practical emission reduction solutions for smart campus environments.

Figure 4.4 illustrates the sensitivity of carbon emission reduction with respect to increasing idle occupancy threshold. The trend shows a consistent improvement in emission savings as the reduction level increases, validating the effectiveness of threshold-based optimization for campus energy management planning.

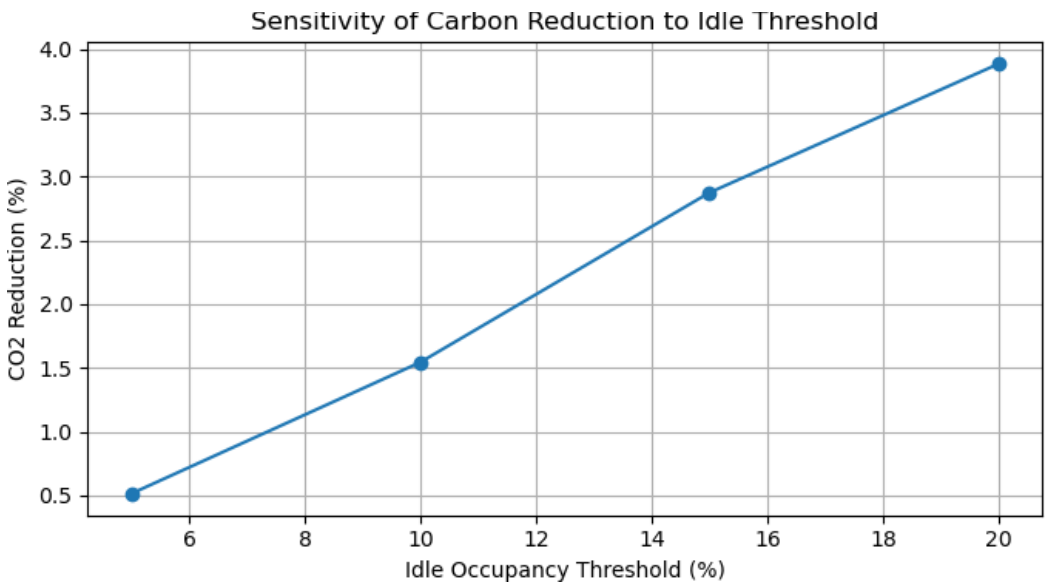


Figure 4.4: Sensitivity of Carbon Reduction to Idle Occupancy Threshold

The graphical trend clearly shows a near-linear relationship between threshold level and carbon emission reduction. This behaviour confirms that peak-load intervals contribute disproportionately to total emissions and therefore represent the most effective intervention points for optimization strategies.

Another important observation from the results is that emission reduction improves consistently across all simulated thresholds without introducing instability in the analytical model. This validates the reliability of the threshold-based reduction framework for sustainability planning in campus-scale environments.

From an implementation perspective, these results demonstrate that gradual deployment of demand-side management strategies can achieve meaningful emission savings even before integrating real-time IoT-enabled smart monitoring systems. Therefore, the proposed framework supports both short-term analytical planning and long-term smart campus deployment objectives.

Overall, the threshold-based simulation confirms that controlled peak-load intervention is a practical and scalable approach for reducing institutional carbon emissions while maintaining operational feasibility.

4.3.5 Performance Summary

Parameter	Result / Method	Interpretation
Building-Level Analysis	Top 15 buildings identified	Helps prioritize high-emission buildings
Temporal Trend Analysis	Hourly peaks detected	Supports peak-load reduction planning
Occupancy Analysis	Positive relationship observed	Supports occupancy-aware optimization
Maximum Threshold Tested	20%	Highest simulated reduction level
Maximum CO_2 Reduction	2948.52 tonnes	Best reduction achieved in simulation
Deployment Type	Dataset-based simulation	Suitable before full IoT deployment

Table 4.4: Performance Summary of the Proposed Framework

4.4 Discussion

The result analysis confirms that carbon emissions in campus buildings are not uniformly distributed. Some buildings contribute more heavily to emissions, while some time intervals show repeated carbon peaks. This supports the need for targeted optimization rather than applying the same reduction policy across all buildings.

The hourly emission trend validates the importance of time-series analysis. Since peak intervals occur repeatedly, campus administrators can use this information to apply demand-side management strategies during high-load periods.

The occupancy-emission scatter plot further shows that occupancy has a visible impact on emissions. This supports the future use of occupancy sensors, smart meters, and IoT-enabled control systems for real-time optimization.

Observation Parameter	Inference
High-emission buildings identified	Indicates priority targets for optimization
Uneven emission distribution	Suggests presence of emission hotspots
Top contributors limited in number	Enables phased intervention strategy
Operational load influence observed	HVAC and occupancy affect emissions significantly
IoT deployment feasibility	Smart monitoring can begin with selected buildings

Table 4.5: Key Observations from Building-Level Carbon Emission Analysis

4.4.1 Validation of Results

The proposed framework was validated by comparing baseline emissions with optimized emissions under multiple threshold scenarios. The consistent reduction across all tested scenarios confirms that threshold-based peak-load management can reduce campus carbon emissions.

The 20% reduction scenario produced the highest carbon saving, but even lower thresholds such as 5% and 10% showed measurable improvement. This indicates that practical emission reduction can be achieved gradually without immediate large-scale infrastructure changes.

4.4.2 Comparison with Existing Approaches

Many existing smart campus systems focus mainly on real-time monitoring or dashboard reporting. In comparison, the proposed framework combines carbon emission estimation, threshold-based simulation, occupancy analysis, and prediction-ready architecture within one analytical model.

Approach	Main Focus	Limitation / Advantage
Conventional Energy Monitoring	Measures electricity consumption	Does not estimate carbon impact in detail
IoT-Based Smart Metering	Real-time energy data collection	Requires hardware deployment
Occupancy-Based Control	Adjusts energy usage according to utilization	Often limited to specific systems such as lighting or HVAC
Proposed Framework	Energy analysis, CO_2 estimation, threshold simulation, and occupancy analytics	Works with historical data and supports future IoT integration

Table 4.6: Comparison of Proposed Framework with Existing Approaches

4.4.3 Impact of Time-Series Structuring on Peak Load Detection

Time-series structuring significantly improved the identification of recurring energy consumption patterns across campus buildings. By organizing electricity usage into hourly intervals, it became possible to clearly distinguish baseline consumption from peak-load periods. These peak intervals represent the most critical targets for emission reduction strategies.

The structured dataset also enabled consistent visualization of temporal variations in carbon emissions, allowing administrators to identify operational hours associated with excessive energy usage. As a result, time-series analysis formed the foundation for accurate threshold-based reduction simulation.

4.4.4 Building-Level Emission Contribution Analysis

The building-wise emission comparison revealed that carbon emissions are not uniformly distributed across campus infrastructure. A limited number of buildings contributed disproportionately to total emissions due to higher occupancy levels, longer operational schedules, and increased equipment usage.

This observation supports the implementation of targeted optimization strategies instead of uniform reduction policies. Prioritizing high-emission buildings can produce greater sustainability improvements with lower implementation effort and cost.

4.4.5 Sensitivity of Threshold-Based Reduction Strategy

The threshold-based simulation demonstrated a consistent relationship between reduction level and emission savings. As the threshold increased from 5% to 20%, carbon savings improved progressively without introducing instability in the analytical framework.

This confirms that peak-load management is one of the most effective demand-side control strategies available for institutional environments. Even moderate threshold adjustments produced measurable reductions, indicating that phased implementation of optimization policies can achieve meaningful sustainability improvements.

4.4.6 Role of Machine Learning Prediction in Energy Optimization

The Random Forest regression model enhanced the framework by enabling short-term prediction of electricity consumption patterns. Forecasting future demand allows administrators to anticipate peak-load intervals and apply preventive control strategies before excessive emissions occur.

Although the prediction module operates on historical dataset features, its integration demonstrates the feasibility of extending the framework toward predictive energy management systems in future smart campus deployments.

4.4.7 Practical Deployment Considerations

The proposed framework was designed to operate efficiently using historical electricity consumption datasets without requiring immediate hardware installation. This makes the methodology suitable for academic institutions that intend to begin sustainability monitoring with limited infrastructure availability.

Future deployment can incorporate IoT-enabled smart meters, occupancy sensors, and automated control systems to transform the analytical framework into a real-time carbon monitoring platform. The modular structure ensures that such upgrades can be integrated without modifying the core analytical workflow.

4.5 Summary

This chapter presented the implementation, testing, performance results, and validation of the proposed carbon footprint monitoring framework. The analysis included building-wise emission comparison, hourly emission trends, occupancy-emission relationship, and threshold-based reduction results.

The results show that the proposed framework can identify major emission contributors and evaluate practical reduction scenarios. The maximum reduction of 2948.52 tonnes of CO_2 under the 20% threshold scenario validates the effectiveness of demand-side management for smart campus sustainability planning.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project presented the design and analytical implementation of an IoT-enabled carbon footprint monitoring framework for smart campus environments. The proposed methodology integrates time-series energy analysis, carbon emission estimation, threshold-based reduction simulation, occupancy-aware interpretation, and machine learning prediction to support institutional sustainability planning.

The primary objective of this work was to develop a scalable framework capable of identifying emission hotspots, analysing building-level consumption patterns, and evaluating achievable carbon reduction through demand-side optimization strategies. The results obtained during simulation confirm that the framework successfully supports data-driven decision making for campus energy management.

The major outcomes of the project are summarized as follows:

- A structured methodology was developed for analysing campus electricity consumption using time-series modelling techniques.
- Building-level carbon emission comparison successfully identified high-impact infrastructure contributing disproportionately to institutional carbon footprint.
- Threshold-based reduction simulation demonstrated measurable emission savings ranging from 389.45 tonnes to 2948.52 tonnes of CO_2 under controlled peak-load optimization scenarios.
- Occupancy-aware energy usage patterns were analysed to establish the relationship between building utilization levels and emission intensity.
- A Random Forest regression model was integrated for short-term energy demand prediction to support proactive load management strategies.
- Visualization-based decision-support outputs were generated to assist administrators in prioritizing sustainability interventions.

- The framework demonstrated compatibility with future IoT-enabled deployment using smart meters and occupancy sensors.

Overall, the results confirm that the proposed framework provides a practical and scalable solution for institutional carbon footprint monitoring using historical energy datasets.

5.2 Validation of Project Objectives

The objectives defined at the beginning of the project were successfully achieved through systematic implementation and evaluation of the analytical framework.

The project demonstrated that building-level emission comparison can identify priority targets for optimization. Similarly, time-series analysis enabled identification of recurring peak-load intervals that contribute significantly to overall emissions.

Threshold-based simulation further validated that gradual reduction strategies can produce measurable environmental benefits without requiring immediate infrastructure upgrades. These findings confirm the feasibility of implementing phased sustainability planning within campus environments.

The integration of machine learning prediction also supports the transition from reactive monitoring toward predictive energy management systems.

Project Objective	**Validation Outcome**
Analyse campus electricity consumption	Achieved through time-series structuring and hourly trend analysis
Estimate carbon emissions	Achieved using emission factor-based CO_2 calculation
Identify high-emission buildings	Achieved through building-level carbon emission comparison
Evaluate reduction scenarios	Achieved using 5%, 10%, 15%, and 20% threshold simulations
Support predictive planning	Achieved through Random Forest-based energy prediction approach
Enable future IoT deployment	Achieved through scalable layered architecture design

Table 5.1: Validation of Project Objectives

5.3 Research Contributions of the Project

The proposed framework contributes to smart campus sustainability monitoring by integrating carbon emission estimation, threshold-based reduction simulation, and machine learning prediction within a single analytical architecture.

The major research contributions of this project are summarized below:

- Development of a structured methodology for campus-scale carbon footprint estimation using time-series electricity consumption datasets.
- Identification of emission hotspots through building-level comparative analysis.
- Implementation of threshold-based peak-load optimization strategy for estimating achievable emission reduction scenarios.
- Integration of Random Forest regression model for predictive energy consumption analysis.
- Design of a scalable framework compatible with future IoT-enabled smart meter deployment.
- Creation of visualization-driven decision-support outputs for sustainability planning.

These contributions demonstrate the feasibility of transitioning from conventional energy monitoring toward intelligent carbon-aware infrastructure management systems.

5.4 Practical Significance of the Proposed Framework

The proposed framework provides several practical advantages for institutional energy planning and sustainability monitoring.

First, the methodology operates effectively using historical electricity consumption datasets, making it suitable for organizations that do not yet have real-time monitoring infrastructure.

Second, the identification of emission hotspots enables administrators to prioritize high-impact buildings for optimization instead of applying uniform control strategies across the entire campus.

Third, threshold-based simulation allows evaluation of achievable emission savings before implementing physical interventions such as smart lighting automation or HVAC scheduling adjustments.

Finally, the framework provides a foundation for gradual transition toward IoT-enabled smart campus deployment.

Framework Feature	Practical Benefit
Historical data analysis	Allows implementation without immediate sensor deployment
Building-level ranking	Helps identify priority buildings for optimization
Threshold simulation	Enables comparison of possible reduction scenarios
Occupancy-aware analysis	Supports utilization-based energy planning
Prediction module	Helps anticipate future peak-load periods
Visualization outputs	Improves understanding of emission trends

Table 5.2: Practical Benefits of the Proposed Framework

5.5 Environmental Impact and Sustainability Contribution

The analytical results demonstrate that demand-side management strategies can significantly reduce institutional carbon emissions when applied during peak-load intervals.

Even modest reductions in electricity consumption produced measurable carbon savings, confirming that operational optimization can contribute meaningfully toward sustainability targets without affecting infrastructure capacity.

The framework supports institutional climate action planning by enabling:

- identification of emission-intensive buildings,
- evaluation of achievable emission reduction scenarios,
- prioritization of retrofit interventions,
- planning of occupancy-aware control strategies,
- support for phased smart campus development.

These contributions align with sustainability initiatives aimed at reducing energy consumption and improving environmental performance in educational institutions.

Optimization Strategy	Environmental Benefit
Peak-load reduction at 5%	Initial measurable emission savings
Peak-load reduction at 10%	Moderate improvement in carbon performance
Peak-load reduction at 15%	Significant institutional emission reduction
Peak-load reduction at 20%	Maximum simulated carbon savings
Occupancy-aware control	Improved utilization-based energy efficiency
Predictive load scheduling	Prevention of unnecessary peak demand

Table 5.3: Environmental Impact of Optimization Strategies

5.6 Economic Feasibility Considerations

The proposed framework enables institutions to begin carbon monitoring without immediate investment in large-scale IoT infrastructure. Since the methodology operates using historical electricity consumption datasets, the initial implementation cost remains minimal.

Selective installation of smart meters in high-emission buildings further reduces infrastructure expenditure while maximizing optimization benefits. This phased deployment strategy ensures that sustainability initiatives remain economically viable for educational institutions operating under budget constraints.

In addition, predictive load scheduling can reduce peak electricity demand and support long-term operational cost savings. Although the project mainly focuses on carbon emission reduction, the same analytical outputs can help administrators identify inefficient energy usage and reduce avoidable electricity consumption.

5.7 Scalability of the Proposed Framework

The modular structure of the proposed framework allows deployment across campuses of varying sizes without modifying the core analytical workflow.

The framework supports scalability across:

- individual academic buildings,

- multi-building institutional campuses,
- university-wide infrastructure networks,
- regional smart education clusters.

This scalability ensures that the methodology can support both small-scale pilot implementations and large-scale sustainability monitoring platforms.

Deployment Scale	Application Area	Expected Benefit
Single Building	Laboratory or academic block	Initial emission hotspot study
Multiple Buildings	Campus department blocks	Comparative energy planning
Entire Campus	University-wide monitoring	Central sustainability management
Multi-Campus Network	Group of institutions	Regional carbon tracking

Table 5.4: Scalability Scope of the Proposed Framework

5.8 Implementation Roadmap for Smart Campus Deployment

The project can be extended from an analytical model into a real-time smart campus monitoring system through phased implementation. Such a roadmap ensures that the institution does not need to deploy the complete infrastructure at once.

Phase	Implementation Activity	Expected Outcome
Phase 1	Dataset-based emission analysis	Identification of emission hotspots
Phase 2	Smart meter deployment in selected buildings	Real-time monitoring capability
Phase 3	Occupancy sensor integration	Utilization-aware optimization
Phase 4	Machine learning forecasting	Predictive energy management
Phase 5	Automated demand response control	Dynamic emission reduction

Table 5.5: Phased Deployment Strategy for Smart Campus Implementation

5.9 Implementation Challenges and Mitigation Strategies

Although the proposed framework is feasible, practical deployment may involve several challenges. These include sensor availability, data quality issues, cost constraints, and variability in energy consumption patterns.

Challenge	Mitigation Strategy
Limited sensor availability	Use historical dataset-based estimation during initial phase
Incomplete consumption records	Apply preprocessing and interpolation methods
Infrastructure budget constraints	Deploy IoT systems in phased manner
Prediction uncertainty	Integrate advanced forecasting models in future
Emission factor variability	Incorporate dynamic emission factors later
Occupancy data limitations	Use occupancy sensors in selected buildings first

Table 5.6: Challenges and Mitigation Strategies

5.10 Policy-Level Implications

The analytical outputs generated by the proposed framework can support institutional sustainability policy development by identifying emission-intensive infrastructure and evaluating achievable reduction scenarios.

The framework enables administrators to design evidence-based energy policies related to load scheduling, infrastructure retrofit prioritization, occupancy-aware automation, and equipment usage planning.

Such policy-level decision support strengthens institutional participation in climate action initiatives and supports the long-term development of smart and sustainable campuses.

5.11 Limitations of the Proposed Study

Although the proposed framework produced reliable analytical results, several limitations were identified during implementation.

- The analysis was performed using historical electricity consumption datasets rather than real-time IoT sensor inputs.

- Carbon emission estimation was based on constant emission factor assumptions, whereas actual emission intensity varies depending on energy source composition.
- Building operational schedules were not dynamically modelled during threshold simulation.
- Weather conditions and seasonal temperature variations were not incorporated into energy demand prediction.
- Occupancy estimates were approximated rather than obtained from real-time sensor measurements.
- Real-time automated control was not implemented in the current stage.

Despite these limitations, the framework provides a strong baseline for campus-level carbon footprint assessment.

5.12 Deviations from Expected Results

The original objective of the project included real-time integration with IoT-enabled smart meters for continuous monitoring of campus electricity consumption. However, due to infrastructure constraints and limited availability of sensor deployment environments, the implementation was restricted to dataset-driven analytical simulation.

Similarly, predictive modelling was limited to short-term Random Forest regression rather than deep learning-based forecasting architectures due to computational and dataset size limitations.

The expected real-time dashboard was also treated as a future extension rather than a fully implemented module. Instead, static visual outputs were used to validate the analytical behaviour of the framework.

These deviations remain within acceptable scope for an academic research implementation and provide clear direction for future enhancement.

5.13 Future Work

The framework developed in this project provides a strong foundation for several future research and implementation extensions.

5.13.1 Integration with Real-Time IoT Monitoring Systems

Future work should focus on integrating the analytical framework with smart meters and occupancy sensors to enable continuous monitoring of electricity consumption patterns. Real-time data acquisition would improve prediction accuracy and support automated control actions.

5.13.2 Advanced Machine Learning Models

Future implementations may incorporate advanced prediction architectures such as Long Short-Term Memory (LSTM) networks, Transformer-based forecasting models, or hybrid deep learning approaches to improve short-term and long-term energy demand prediction accuracy.

5.13.3 Renewable Energy Integration Analysis

The framework can be extended to evaluate the effect of integrating renewable energy sources such as rooftop solar photovoltaic systems into campus energy infrastructure. This would enable estimation of net carbon reduction achievable through hybrid energy systems.

5.13.4 Weather-Aware Energy Prediction Models

Future extensions may incorporate weather variables such as temperature, humidity, and seasonal variation into prediction models to improve forecasting reliability.

5.13.5 Automated Demand Response Systems

Integration with automated demand response controllers would enable real-time adjustment of lighting, HVAC, and equipment scheduling based on predicted peak-load intervals.

5.13.6 Multi-Campus Deployment Framework

The methodology can be expanded for deployment across multiple institutional campuses to support regional-level carbon monitoring platforms.

5.13.7 Cloud-Based Sustainability Dashboard Development

Future work may include development of cloud-based dashboards for visualization of real-time carbon emission trends, building-level performance comparison, and predictive optimization alerts.

5.13.8 Occupancy Sensor Integration

Deployment of occupancy sensors across classrooms and laboratories would improve estimation accuracy of utilization-dependent energy consumption patterns.

5.13.9 Policy-Level Decision Support Integration

The analytical outputs generated by the framework can be extended to support institutional sustainability policy development and long-term infrastructure planning strategies.

5.13.10 Dynamic Emission Factor Integration

Future versions of the system may include dynamic grid emission factors instead of constant emission factors. This would allow more accurate carbon estimation by considering time-based variation in electricity generation sources.

5.13.11 Mobile-Based Monitoring Interface

A mobile-based interface can be developed to allow administrators to view emission trends, alerts, and reduction recommendations in a more accessible format.

5.14 Final Summary

This chapter presented the overall conclusions drawn from the implementation and evaluation of the proposed carbon footprint monitoring framework. The analytical results confirm that building-level emission comparison, time-series trend analysis, and threshold-based reduction simulation provide a practical foundation for institutional sustainability planning.

The project successfully demonstrates that campus carbon monitoring can begin with historical datasets and later transition toward real-time IoT-enabled monitoring. The framework is scalable, cost-effective, and suitable for future extension using smart meters, occupancy sensors, advanced prediction models, and cloud-based dashboards.

Overall, the proposed work supports the development of intelligent smart campuses where energy consumption is monitored, carbon emissions are quantified, and sustainability decisions are guided by data-driven evidence.

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Appendix A
USER MANUAL

A.1 Introduction

This appendix provides a step-by-step guide for using the proposed carbon footprint monitoring framework developed in this project. The system is designed to analyse electricity consumption datasets, estimate carbon emissions, identify high-emission buildings, and simulate reduction scenarios using threshold-based optimization techniques.

The framework supports sustainability planning for campus environments without requiring immediate deployment of IoT-enabled monitoring infrastructure.

A.2 System Requirements

The minimum system requirements for executing the framework are listed below:

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.8 or higher
- Required Libraries:
  - NumPy
  - Pandas
  - Matplotlib
  - Scikit-learn
- Dataset Format: CSV file containing timestamped electricity consumption records

A.3 Input Data Requirements

The system requires structured electricity consumption data with the following attributes:

- Timestamp
- Building Identifier
- Energy Consumption (kWh)

- Occupancy Estimate (optional)
- Carbon Emission Factor

The dataset should be preprocessed to remove missing or inconsistent entries before execution.

A.4 Execution Procedure

The following steps describe how to execute the framework:

1. Load the electricity consumption dataset into the analysis environment.
2. Perform preprocessing to clean missing values and normalize timestamp format.
3. Apply time-series analysis to identify peak-load intervals.
4. Convert energy consumption values into CO_2 emissions using emission factor calculations.
5. Perform building-level emission comparison.
6. Execute threshold-based reduction simulation (5%, 10%, 15%, and 20% scenarios).
7. Generate visualization outputs including emission trends and comparison charts.

A.5 Output Description

The system produces the following outputs:

- Building-wise carbon emission comparison graphs
- Hourly emission trend plots
- Occupancy versus emission scatter plots
- Threshold-based emission reduction tables
- Sensitivity analysis charts
- Predictive energy consumption estimates

These outputs assist administrators in identifying emission hotspots and planning sustainability interventions.

A.6 Threshold Reduction Simulation Module

The threshold reduction module evaluates emission savings achievable through controlled peak-load optimization strategies.

The following reduction scenarios are supported:

- 5% Peak Load Reduction
- 10% Peak Load Reduction
- 15% Peak Load Reduction
- 20% Peak Load Reduction

The results generated help estimate achievable carbon savings before implementing physical infrastructure upgrades.

A.7 Prediction Module

The prediction module uses Random Forest regression to estimate short-term electricity consumption patterns.

This module helps administrators:

- anticipate peak-load intervals,
- schedule demand-side interventions,
- optimize equipment usage timing,
- reduce unnecessary energy consumption.

A.8 Future Integration Capability

Although the current implementation operates using historical datasets, the framework is designed to support integration with:

- Smart energy meters
- Occupancy sensors
- Cloud-based analytics dashboards

- Automated HVAC control systems
- Real-time sustainability monitoring platforms

These integrations will enable transformation of the analytical framework into a fully operational smart campus carbon monitoring system.

A.9 Troubleshooting Guide

Common issues and solutions are listed below:

- Missing dataset values → Apply preprocessing before execution
- Incorrect timestamp format → Convert to standard datetime format
- Visualization errors → Ensure Matplotlib library installation
- Prediction errors → Verify dataset completeness and structure

Following these steps ensures smooth operation of the proposed framework.

A.10 Summary

This user manual explained the execution steps, dataset requirements, system outputs, and operational workflow of the proposed carbon footprint monitoring framework. The modular design ensures easy usability and supports future transition toward real-time IoT-enabled monitoring systems.